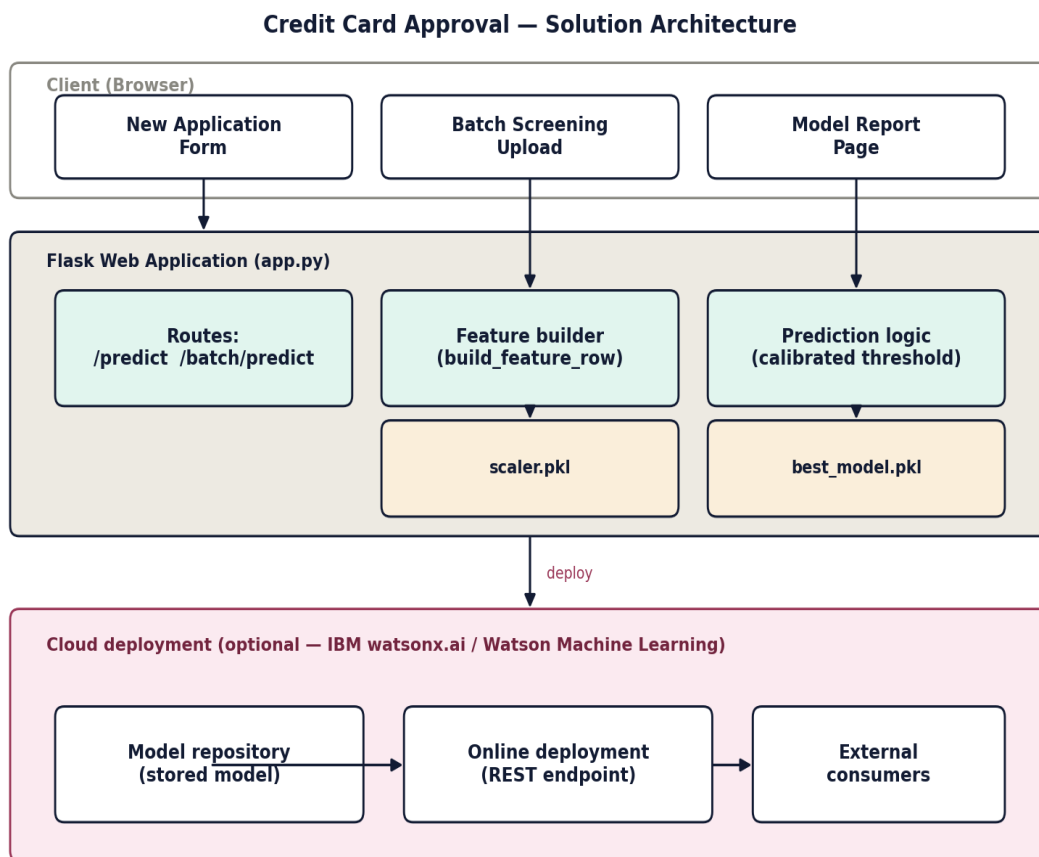


Project Design Phase

Solution Architecture

Date	06 July 2026
Team ID	
Project Name	Ledgerline – Credit Card Approval Prediction System
Maximum Marks	4 Marks

Solution Architecture Diagram:



The architecture has four layers: a browser-based Presentation/Client Layer (new application form, batch CSV upload, model report page); an API Gateway/Routing layer (Flask routes: `/predict`, `/batch/predict`, `/api/predict`, `/about`); a Core Logic Service (feature engineering and calibrated ML model inference using `scaler.pkl` and `best_model.pkl`); and a Data/Storage layer (`application_record.csv`, `credit_record.csv`, and the saved model artifacts in `models/`). An optional External API Integration layer connects to IBM watsonx.ai for cloud-hosted scoring.

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Browser-based pages for entering a single application, uploading a batch CSV, and viewing the model report.	HTML, CSS, Jinja2 templates
API Gateway / Routing	Flask routes that receive requests and dispatch to the right handler.	Flask
Core Logic Service	Builds the exact feature vector used in training, scales it, and scores it with the calibrated model.	Python, pandas, scikit-learn, XGBoost
External API Integrations	Optional cloud-hosted scoring endpoint for the same model.	IBM watsonx.ai (Watson Machine Learning)
Data / Storage Layer	Source data and saved model artifacts.	CSV files, joblib-serialized model & scaler